

Chapter 5 – Preparing Ubuntu Server for Wazuh Installation

Purpose

Before installing Wazuh, prepare Ubuntu Server to ensure it is updated, properly configured, and ready to host Wazuh services.

1. Verify Ubuntu Version

Command

```
lsb_release -a
```

Purpose

- Displays Ubuntu version information.
 - Verifies compatibility with Wazuh.
 - Useful during troubleshooting.
-

2. Update the Package List

Command

```
sudo apt update
```

Purpose

- Refreshes the package database.
- Retrieves the latest package information from Ubuntu repositories.
- **Does not install updates.**

Easy Memory

Update = Check for new software versions.

3. Upgrade Installed Packages

Command

```
sudo apt upgrade -y
```

Purpose

- Installs available software updates.
- Applies security patches and bug fixes.
- Keeps the operating system secure before installing Wazuh.

Easy Memory

Upgrade = Install the updates.

4. Check Hostname

Command

```
hostnamectl
```

Purpose

- Displays the computer's hostname.
 - Shows OS version, kernel, and architecture.
-

5. Change Hostname

Command

```
sudo hostnamectl set-hostname wazuh-server
```

Purpose

Assigns a meaningful name to the server.

Example:

wazuh-server

instead of

ubuntu

Why Use a Meaningful Hostname?

Makes servers easier to identify.

Example:

database-server

web-server

wazuh-server

backup-server

instead of

ubuntu01

ubuntu02

ubuntu03

6. Check Network Configuration

Command

ip addr

Purpose

Displays:

- Network Interfaces
- IP Address
- Interface Status

Used to verify network connectivity.

7. Test Internet Connection

Command

ping google.com

Purpose

Checks whether Ubuntu can access the Internet.

Stop using:

Ctrl + C

8. Static IP Address

Why do we need it?

DHCP assigns dynamic IP addresses.

Example:

Today

192.168.56.101

Tomorrow

192.168.56.120

If the IP changes, Wazuh Agents may lose connection to the Wazuh Manager.

A Static IP remains the same.

Example:

192.168.56.10

9. What is Netplan?

Netplan is Ubuntu's network configuration utility.

It is used to configure:

- Static IP Address

- DHCP
- DNS Servers
- Gateway
- Network Interfaces

Configuration files are stored in:

/etc/netplan/

10. What is YAML?

YAML stands for:

YAML Ain't Markup Language

It is a **human-readable configuration file format** used to store settings.

Ubuntu uses YAML files to configure networking through Netplan.

Why does Ubuntu use YAML?

Because YAML is:

- Easy to read
- Easy to write
- Structured
- Used by many modern tools such as:
 - Netplan
 - Docker Compose
 - Kubernetes
 - Ansible
 - GitHub Actions

Example YAML File

```
network:
  version: 2
  ethernets:
    enp0s3:
      dhcp4: true
```

Understanding the YAML Structure

network:
Starts the network configuration section.

version: 2
Specifies the Netplan configuration version.

ethernets:
Defines Ethernet network interfaces.

enp0s3:
The name of the network interface.

dhcp4: true
Enables IPv4 address assignment using DHCP.

Important YAML Rule

YAML uses **indentation (spaces)** instead of braces `{}` or brackets `[]`.

Correct:

```
network:  
  version: 2
```

Incorrect:

```
network:  
version: 2
```

Even one missing space can cause configuration errors.

11. Install Required Utilities

Command

```
sudo apt install curl unzip tar -y
```

Purpose

Installs utilities required during the Wazuh installation.

Packages

curl

Downloads files from websites.

unzip

Extracts .zip files.

tar

Extracts .tar and .tar.gz archives.

Chapter 5 Summary

- Verify Ubuntu version.
- Update package list.
- Upgrade installed packages.
- Configure a meaningful hostname.
- Verify network connectivity.
- Understand why a static IP is required.
- Learn Netplan.
- Understand YAML configuration files.
- Install required utilities.

Quick Revision

Command	Purpose
lsb_release -a	Check Ubuntu version
sudo apt update	Refresh package list
sudo apt upgrade -y	Install updates
hostnamectl	View hostname
sudo hostnamectl set-hostname wazuh-server	Change hostname
ip addr	View IP address
ping google.com	Test Internet
sudo apt install curl unzip tar -y	Install utilities

One important clarification about YAML

Many beginners think **YAML is a programming language**. It is **not**.

-  YAML does **not** execute code.
-  You don't write programs in YAML.
-  YAML is a **configuration language** used to describe settings in a structured, human-readable way.

For example, when you configure a static IP in Ubuntu, you're not writing a program—you are simply telling Netplan **how the network should be configured**. Netplan reads the YAML file and applies those settings to the system.